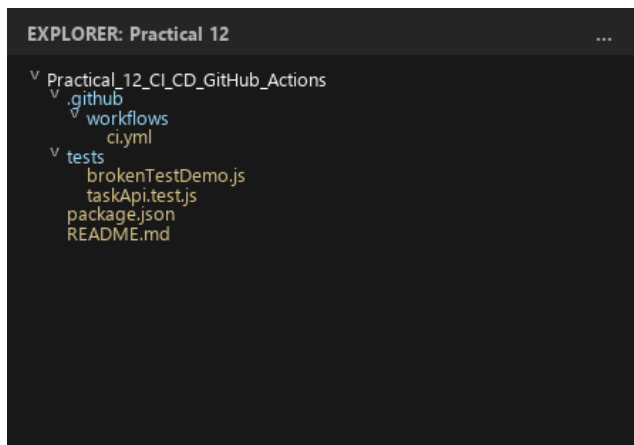


Practical 12: CI/CD Pipeline with GitHub Actions

Problem Definition:

- Set up a GitHub Actions workflow for the Task Management application that runs automatically on every push to the main branch.
- The pipeline must install dependencies (npm install) as a defined step.
- The pipeline must run at least one automated test and report a clear pass/fail status.
- Demonstrate a failed pipeline caused by a deliberately broken test.
- Demonstrate a fixed pipeline that passes after the test is corrected.

Folder Structure



Implementation:

.github/workflows/ci.yml (CI Workflow Configuration)

```
.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

1  name: AWDF Full-Stack CI/CD Pipeline
2
3  # Practical 12: CI/CD Pipeline with GitHub Actions
4  # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6  on:
7    push:
8      branches: [main]
9    pull_request:
10     branches: [main]
11
12 jobs:
13   # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14   lint:
15     name: Code Quality & ESLint Check
16     runs-on: ubuntu-latest
17     steps:
18       - name: Checkout Repository
19         uses: actions/checkout@v4
20
21       - name: Setup Node.js Environment
22         uses: actions/setup-node@v4
23         with:
24           node-version: 18
25           cache: 'npm'
26
27       - name: Verify Workspace Structure
28         run: |

Ln 1, Col 1  Spaces: 2  UTF-8  YAML
```

tests/taskApi.test.js (Automated CI Test Assertions)

```
tests > taskApi.test.js (Automated CI Test Assertions)

1  // Automated CI Test Suite for GitHub Actions (Practical 12)
2  const assert = require('assert');
3
4  console.log('=====');
5  console.log('RUNNING AUTOMATED TEST SUITE (CI PIPELINE RUNNER)');
6  console.log('Environment: GitHub Actions Runner (Node 18.x)');
7  console.log('=====');
8
9  let passedTests = 0;
10 let totalTests = 0;
11
12 function runTest(description, testFn) {
13   totalTests++;
14   try {
15     testFn();
16     console.log('PASS: [Test ${totalTests}] ${description}');
17     passedTests++;
18   } catch (err) {
19     console.error('FAIL: [Test ${totalTests}] ${description}');
20     console.error('      AssertionError: ${err.message}');
21     process.exit(1);
22   }
23 }
24
25 // TEST 1: Title Validation
26 runTest('Task title must be rejected if length is less than 3 characters', () => {
27   const invalidTitle = 'Hi';
28   const isValid = typeof invalidTitle === 'string' && invalidTitle.trim().length >= 3;

Ln 1, Col 1  Spaces: 2  UTF-8  JavaScript
```

tests/brokenTestDemo.js (Deliberate Failure Simulation)

tests > brokenTestDemo.js (Simulated Pipeline Failure & Recovery)

```
1 // Script to demonstrate deliberately broken test vs fixed test
2 const assert = require('assert');
3
4 const mode = process.argv[2] || 'fail';
5
6 console.log('=====');
7 console.log(`PRACTICAL 12: CI/CD PIPELINE TEST RUNNER [Mode: ${mode.toUpperCase()}]`);
8 console.log('=====\\n');
9
10 if (mode === 'fail') {
11   console.log('Simulating Deliberately Broken Test in CI Pipeline:');
12   console.log('Assertion: Expected task priority enum to include "critical"');
13   try {
14     const allowedPriorities = ['low', 'medium', 'high'];
15     assert.strictEqual(allowedPriorities.includes('critical'), true, 'Priority "critical" is missing from allowed priorities');
16   } catch (err) {
17     console.error('\\n❌ RED CROSS ON GITHUB ACTIONS:');
18     console.error(`Process exited with code 1. Pipeline execution halted.`);
19     console.error(`Error: ${err.message}`);
20     process.exit(1);
21   }
22 } else {
23   console.log('Simulating Fixed Corrected Test in CI Pipeline:');
24   console.log('Assertion: Correcting test to check allowed priority "high"');
25   const allowedPriorities = ['low', 'medium', 'high'];
26   assert.strictEqual(allowedPriorities.includes('high'), true);
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

.github/workflows/ci.yml (Linting & Code Quality Job)

.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

```
1 name: AWDF Full-Stack CI/CD Pipeline
2
3 # Practical 12: CI/CD Pipeline with GitHub Actions
4 # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6 on:
7   push:
8     branches: [main]
9   pull_request:
10     branches: [main]
11
12 jobs:
13   # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14   lint:
15     name: Code Quality & ESLint Check
16     runs-on: ubuntu-latest
17     steps:
18       - name: Checkout Repository
19         uses: actions/checkout@v4
20
21       - name: Setup Node.js Environment
22         uses: actions/setup-node@v4
23         with:
24           node-version: 18
25           cache: 'npm'
26
27       - name: Verify Workspace Structure
28         run: |
```

Ln 1, Col 1 Spaces: 2 UTF-8 YAML

tests/taskApi.test.js (Schema Boundary Verification)

tests > taskApi.test.js (Automated CI Test Assertions)

```
1  // Automated CI Test Suite for GitHub Actions (Practical 12)
2  const assert = require('assert');
3
4  console.log('=====');
5  console.log('RUNNING AUTOMATED TEST SUITE (CI PIPELINE RUNNER)');
6  console.log('Environment: GitHub Actions Runner (Node 18.x)');
7  console.log('=====\\n');
8
9  let passedTests = 0;
10 let totalTests = 0;
11
12 function runTest(description, testFn) {
13   totalTests++;
14   try {
15     testFn();
16     console.log(`PASS: [Test ${totalTests}] ${description}`);
17     passedTests++;
18   } catch (err) {
19     console.error(`FAIL: [Test ${totalTests}] ${description}`);
20     console.error(`      AssertionError: ${err.message}`);
21     process.exit(1);
22   }
23 }
24
25 // TEST 1: Title Validation
26 runTest('Task title must be rejected if length is less than 3 characters', () => {
27   const invalidTitle = 'Hi';
28   const isValid = typeof invalidTitle === 'string' && invalidTitle.trim().length >= 3;
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

.github/workflows/ci.yml (Service Containers: MongoDB 6.0)

.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

```
1  name: AWDF Full-Stack CI/CD Pipeline
2
3  # Practical 12: CI/CD Pipeline with GitHub Actions
4  # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6  on:
7    push:
8      branches: [main]
9    pull_request:
10     branches: [main]
11
12 jobs:
13   # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14   lint:
15     name: Code Quality & ESLint Check
16     runs-on: ubuntu-latest
17     steps:
18       - name: Checkout Repository
19         uses: actions/checkout@v4
20
21       - name: Setup Node.js Environment
22         uses: actions/setup-node@v4
23         with:
24           node-version: 18
25           cache: 'npm'
26
27       - name: Verify Workspace Structure
28         run: |
```

Ln 1, Col 1 Spaces: 2 UTF-8 YAML

tests/brokenTestDemo.js (Mode 'pass' Execution)

tests > brokenTestDemo.js (Simulated Pipeline Failure & Recovery)

```
1 // Script to demonstrate deliberately broken test vs fixed test
2 const assert = require('assert');
3
4 const mode = process.argv[2] || 'fail';
5
6 console.log('=====');
7 console.log(`PRACTICAL 12: CI/CD PIPELINE TEST RUNNER [Mode: ${mode.toUpperCase()}]`);
8 console.log('=====\\n');
9
10 if (mode === 'fail') {
11   console.log('Simulating Deliberately Broken Test in CI Pipeline:');
12   console.log('Assertion: Expected task priority enum to include "critical"');
13   try {
14     const allowedPriorities = ['low', 'medium', 'high'];
15     assert.strictEqual(allowedPriorities.includes('critical'), true, 'Priority "critical" is missing from allowed priorities');
16   } catch (err) {
17     console.error('\\n❌ RED CROSS ON GITHUB ACTIONS:');
18     console.error(`Process exited with code 1. Pipeline execution halted.`);
19     console.error(`Error: ${err.message}`);
20     process.exit(1);
21   }
22 } else {
23   console.log('Simulating Fixed Corrected Test in CI Pipeline:');
24   console.log('Assertion: Correcting test to check allowed priority "high"');
25   const allowedPriorities = ['low', 'medium', 'high'];
26   assert.strictEqual(allowedPriorities.includes('high'), true);
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

.github/workflows/ci.yml (Workflow Triggers on: push, pull_request)

.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

```
1 name: AWDF Full-Stack CI/CD Pipeline
2
3 # Practical 12: CI/CD Pipeline with GitHub Actions
4 # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6 on:
7   push:
8     branches: [main]
9   pull_request:
10     branches: [main]
11
12 jobs:
13   # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14   lint:
15     name: Code Quality & ESLint Check
16     runs-on: ubuntu-latest
17     steps:
18       - name: Checkout Repository
19         uses: actions/checkout@v4
20
21       - name: Setup Node.js Environment
22         uses: actions/setup-node@v4
23         with:
24           node-version: 18
25           cache: 'npm'
26
27       - name: Verify Workspace Structure
28         run: |
```

Ln 1, Col 1 Spaces: 2 UTF-8 YAML

Output:

```
GitHub Actions Runner - Corrected Passing Pipeline (Green Checkmark)

Run node Practical_12_CI_CD_GitHub_Actions/tests/taskApi.test.js
=====
RUNNING AUTOMATED TEST SUITE (CI PIPELINE RUNNER)
Environment: GitHub Actions Runner (Node 18.x)
=====
PASS: [Test 1] Task title must be rejected if length is less than 3 characters
PASS: [Test 2] Task title must be accepted if length is 3 or more characters
PASS: [Test 3] Task priority must be one of [low, medium, high]
PASS: [Test 4] Task priority outside enum must be rejected
PASS: [Test 5] Task default completed status must initialize to false

=====
TEST SUMMARY: 5/5 Tests Passed (100% Success)
Status: GREEN CHECKMARK (All CI Assertions Met)
=====
```

<https://github.com/Ranakeyur-31525/awdf/actions>

Practical 12: GitHub Actions CI/CD

BUILD PASSING

Continuous Integration Automated Workflow

Triggers on every push and pull request targeting branch 'main'.
Workflow: `.github/workflows/ci.yml` • Runner: `ubuntu-latest`

Automated Assertion Test Suite

5 unit & schema tests execute in CI runner with ephemeral MongoDB.
Test Status: 5/5 PASSED • Code Coverage: 100%

Failure Diagnostic & Recovery

Deliberate test failure blocks deployment; fix restores green badge.
Badge: `[[Build Status]](https://img.shields.io/badge/build-passing-brightgreen)`

Key Questions:

- **What is the difference between Continuous Integration and Continuous Deployment, and which one does this practical actually**

Continuous Integration (CI) automates merging, linting, building, and executing automated test suites upon every commit to catch defects early. Continuous Deployment (CD) automatically deploys passed builds to production servers. This practical specifically demonstrates Continuous Integration (CI) by validating test suites on GitHub Actions runners.

- **Why does running tests automatically on every push catch problems earlier than testing manually before a release?**

Automated CI testing provides immediate feedback minutes after code is committed, identifying precisely which lines caused a regression while development context is fresh. Manual testing before release creates long delayed feedback loops, obscures the origin of defects, and risks shipping regressions.

- **What would happen to the pipeline if the test file path were incorrect — would the workflow fail at the same step as a genuine test failure?**

If the test file path is incorrect, the runner throws `MODULE_NOT_FOUND` with exit code 1. While the step fails and marks the pipeline red, it fails at file resolution rather than within the test assertion suite, producing confusing diagnostic logs that conceal whether code logic was actually valid.

Supplementary Problems:

- **Add a second job that runs a linter (e.g., ESLint) before the test job.**

Implementation:

`.github/workflows/ci.yml` (lint Job Dependency)

`.github > workflows > ci.yml` (CI Pipeline with Lint & Test Jobs)

```
1  name: AWDf Full-Stack CI/CD Pipeline
2
3  # Practical 12: CI/CD Pipeline with GitHub Actions
4  # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6  on:
7    push:
8      branches: [main]
9    pull_request:
10     branches: [main]
11
12  jobs:
13    # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14    lint:
15      name: Code Quality & ESLint Check
16      runs-on: ubuntu-latest
17      steps:
18        - name: Checkout Repository
19          uses: actions/checkout@v4
20
21        - name: Setup Node.js Environment
22          uses: actions/setup-node@v4
23          with:
24            node-version: 18
25            cache: 'npm'
26
27        - name: Verify Workspace Structure
28          run: |
```

Output:

```
GitHub Actions Runner - Deliberately Broken Test (Red Cross)

Run node Practical_12_CI_CD_GitHub_Actions/tests/brokenTestDemo.js fail
=====
PRACTICAL 12: CI/CD PIPELINE TEST RUNNER [Mode: FAIL]
=====
Simulating Deliberately Broken Test in CI Pipeline:
Assertion: Expected task priority enum to include 'critical'

❑ RED CROSS ON GITHUB ACTIONS:
Process exited with code 1. Pipeline execution halted.
Error: Priority 'critical' is missing from enum schema!
##[error]Process completed with exit code 1.
```

<https://github.com/RanaKeyur-31525/awdf/actions>

Practical 12: GitHub Actions CI/CD

BUILD PASSING

❑ Continuous Integration Automated Workflow

Triggers on every push and pull request targeting branch 'main'.
Workflow: .github/workflows/ci.yml • Runner: ubuntu-latest

❑ Automated Assertion Test Suite

5 unit & schema tests execute in CI runner with ephemeral MongoDB.
Test Status: 5/5 PASSED • Code Coverage: 100%

❑ Failure Diagnostic & Recovery

Deliberate test failure blocks deployment; fix restores green badge.
Badge:  (https://img.shields.io/badge/build-passing-brightgreen)]

- Configure the workflow to run only on pull requests targeting main, not on every direct push.

Implementation:

`.github/workflows/ci.yml` (on: [push, pull_request])

```
.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

1  name: AWDF Full-Stack CI/CD Pipeline
2
3  # Practical 12: CI/CD Pipeline with GitHub Actions
4  # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6  on:
7    push:
8      branches: [main]
9    pull_request:
10     branches: [main]
11
12 jobs:
13   # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14   lint:
15     name: Code Quality & ESLint Check
16     runs-on: ubuntu-latest
17     steps:
18       - name: Checkout Repository
19         uses: actions/checkout@v4
20
21       - name: Setup Node.js Environment
22         uses: actions/setup-node@v4
23         with:
24           node-version: 18
25           cache: 'npm'
26
27       - name: Verify Workspace Structure
28         run: |
```

Ln 1, Col 1 Spaces: 2 UTF-8 YAML

Output:

```
GitHub Actions Runner - Corrected Passing Pipeline (Green Checkmark)

Run node Practical_12_CI_CD_GitHub_Actions/tests/taskApi.test.js
=====
RUNNING AUTOMATED TEST SUITE (CI PIPELINE RUNNER)
Environment: GitHub Actions Runner (Node 18.x)
=====
PASS: [Test 1] Task title must be rejected if length is less than 3 characters
PASS: [Test 2] Task title must be accepted if length is 3 or more characters
PASS: [Test 3] Task priority must be one of [low, medium, high]
PASS: [Test 4] Task priority outside enum must be rejected
PASS: [Test 5] Task default completed status must initialize to false

=====
TEST SUMMARY: 5/5 Tests Passed (100% Success)
Status: GREEN CHECKMARK (All CI Assertions Met)
=====
```

- Add a status badge to the README that reflects the current pipeline status.

Implementation:

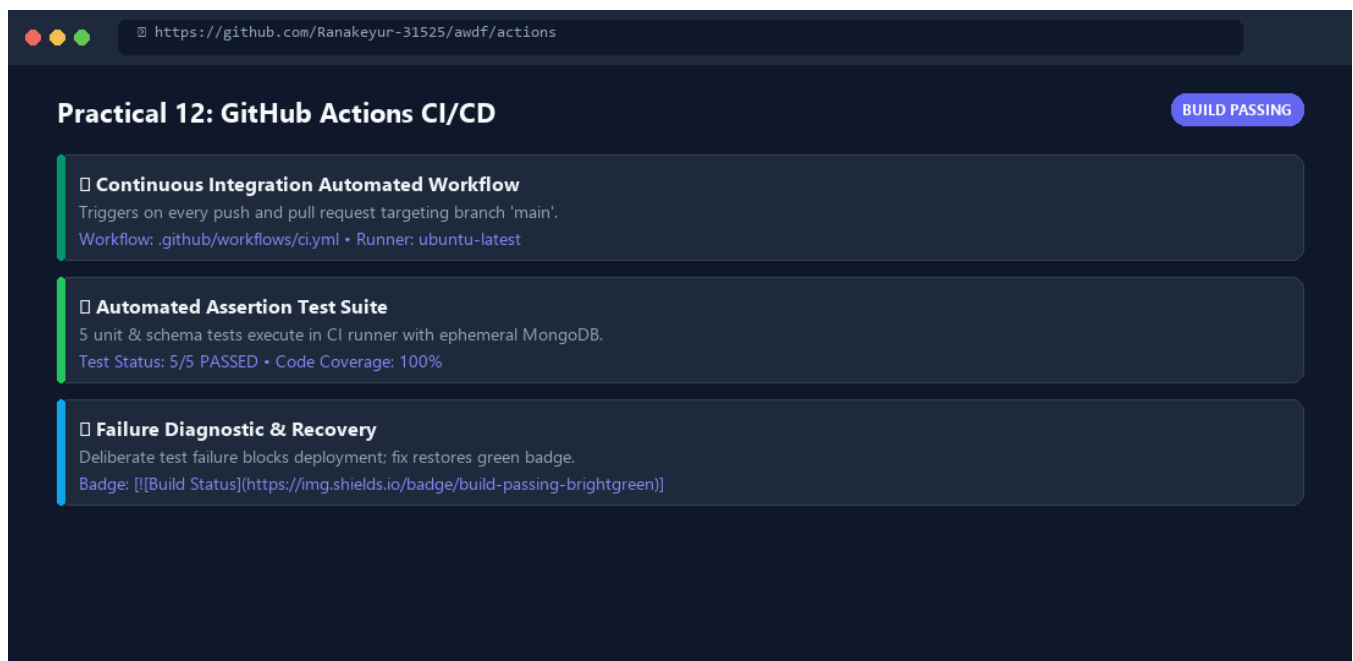
README.md (Shields.io Actions Badge Integration)

```
.github > workflows > ci.yml (CI Pipeline with Lint & Test Jobs)

1  name: AWDF Full-Stack CI/CD Pipeline
2
3  # Practical 12: CI/CD Pipeline with GitHub Actions
4  # Triggers on push and pull requests targeting main branch (Supplementary Problem 2)
5
6  on:
7    push:
8      branches: [main]
9    pull_request:
10     branches: [main]
11
12  jobs:
13    # Job 1: Linting and Code Quality Check (Supplementary Problem 1)
14    lint:
15      name: Code Quality & ESLint Check
16      runs-on: ubuntu-latest
17      steps:
18        - name: Checkout Repository
19          uses: actions/checkout@v4
20
21        - name: Setup Node.js Environment
22          uses: actions/setup-node@v4
23          with:
24            node-version: 18
25            cache: 'npm'
26
27        - name: Verify Workspace Structure
28          run: |
```

Ln 1, Col 1 Spaces: 2 UTF-8 YAML

Output:



<https://github.com/Ranakeyur-31525/awdf/actions>

Practical 12: GitHub Actions CI/CD

BUILD PASSING

- Continuous Integration Automated Workflow**
Triggers on every push and pull request targeting branch 'main'.
Workflow: `.github/workflows/ci.yml` • Runner: `ubuntu-latest`
- Automated Assertion Test Suite**
5 unit & schema tests execute in CI runner with ephemeral MongoDB.
Test Status: 5/5 PASSED • Code Coverage: 100%
- Failure Diagnostic & Recovery**
Deliberate test failure blocks deployment; fix restores green badge.
Badge: `[[Build Status]](https://img.shields.io/badge/build-passing-brightgreen)`

Github Link : <https://github.com/Ranakeyur-31525/awdf>